

Part B:

Question No 2:

1. Problem Analysis Chart (PAC):

Input	Processing	Output	Decision
N (Total reqs)	• Set $current_floor = 0$	• Direction message ("Moving up", "Moving down")	• $target_floor > current_floor$ $\Rightarrow$ Print "Moving up"
• $target_floor$ (Required floor per situation)	• loop N times to process each request	• "Doors Opening" • $current_floor$ update notification	• $target_floor < current_floor$ $\Rightarrow$ Print "Moving down"
	• Compare $target_floor$ with $current_floor$		• $target_floor = current_floor$ $\Rightarrow$ "Doors Opening"
	• Print appropriate direction message		
	• Update $current_floor$ to $target_floor$		

2. IPO Chart.

Input	Processing	Output
• N	1. Initialize current floor = 0	• Direction Message
• Target floor	2. Pro-pt user for no of reqs N	• Current floor
	3. For each req 1 to N	
	a) Read Target floor	
	b) if Target floor > current floor print "Moving up" ; $\text{current floor} = \text{current floor} + 1$	
	c) Else if Target floor < current floor print "Moving Down" ; $\text{current floor} = \text{current floor} - 1$	
	d) Else print "Doors Opening"	
	e) Set current floor = target floor	
	f) print updated current floor	

art 3. Algorithm:

1. Start

2. initialize current floor = 0

3. Input total number of floor requests, N .

4. Set counter $i = 1$

5. Repeat while $i \leq N$

- input target floor

- if target floor $>$ current floor

- Display "Moving up"

- Else if target floor $<$ current floor

- Display "Moving Down"

- Else

- Display "Doors Opening"

- Update

current floor = target floor

- Display Updated current floor

- Increment $i = i + 1$

6) END

4-Pseudocode:

Begin

DECLARE N , current floor, counter, target floor^{as int}

SET current floor = 0

Print "Enter total no. of requests."

READ N

FOR counter = 1 to N

Print "Enter target floor"

READ target floor

IF (target floor > current floor)

Print "Moving up"

Else If (target floor < current floor)

Print "Moving Down"

Else

Print "Doors Opening"

END FOR

END

s. flow chart:

